mnirs v0.8.0 Muscle Near-Infrared Spectroscopy Processing and Analysis

online app

Read, process, and analyse data from muscle near-infrared spectroscopy (mNIRS) devices. Import raw data from file and return time-series data and metadata. Standardised methods for cleaning, filtering, transforming, and analysing mNIRS data. Custom plot theme and colour palette.

Workflow

```
├── read_mnirs()
│   ├── create_mnirs_data()
│   └── example_mnirs()
├── resample_mnirs()
├── replace_mnirs()
│   ├── replace_invalid()
│   ├── replace_outliers()
│   └── replace_missing()
├── filter_mnirs()
│   ├── filter_moving_average()
│   └── filter_butterworth()
├── shift_mnirs()
├── rescale_mnirs()
├── correct_blood_volume()
│   ├── theme_mnirs()
│   ├── palette_mnirs()
│   ├── scale_colour_mnirs()
│   ├── scale_fill_mnirs()
│   ├── format_hmmss()
│   └── breaks_timespan()
├── extract_intervals()
│   ├── by_time()
│   ├── by_label()
│   ├── by_lap()
│   └── by_sample()
├── analyse_kinetics()
│   ├── response_time()
│   ├── peak_slope()
│   ├── monoexponential()
│   ├── exponential_drift()
│   ├── biexponential()
│   └── logistic()
│   └── gompertz()
│   └── sigmoidal_drift()
└── plot.mnirs()
```

Read

`read_mnirs()` — Import mnirs data from exported files

Package-level functions

`example_mnirs()` — Paths to included demo files

`create_mnirs_data()` — Add class "*mnirs*" metadata to data frame

Resample

`resample_mnirs()` — Resample to a regular time grid

Clean

`replace_mnirs()` — Replace outliers, invalid, and missing data

Vector-level functions

`replace_outliers()` — Replace local outliers

`replace_invalid()` — Replace specific invalid values

`replace_missing()` — Interpolate over *NA* values

Filter

`filter_mnirs()` — Apply digital filter to mNIRS channels

Vector-level functions

`filter_butterworth()` — Apply a Butterworth digital filter

`filter_moving_average()` — Apply moving average smoothing

Transform

`shift_mnirs()` — Shift channels to a reference value

`rescale_mnirs()` — Rescale channels amplitude / range

`correct_blood_volume()` — Normalise total[haem] changes

Extract

`extract_intervals()` — Detect events and extract intervals

Interval boundary helpers

`by_time()`, `by_label()`, `by_lap()`, `by_sample()`

— Boundaries by time value, event string, lap integer, or row index

Analyse

`analyse_kinetics()` — Analyse response kinetics

Vector-level functions

`response_time()` — Compute fractional response time

`peak_slope()` — Find peak linear slope

`monoexponential()` — Monoexponential curve fitting

`exponential_drift()` — Two-phase exponential-linear

`biexponential()` — Two-phase overlapping exponentials

`logistic()`, `gompertz()` — Sigmoidal functions

`sigmoidal_drift()` — Two-phase sigmoidal-linear

Plot

`plot()` — Plot a data frame of class "*mnirs*" or "*mnirs_kinetics*"

`theme_mnirs()` — ggplot2 theme for mnirs plots

`palette_mnirs()` — Access the mnirs colour palette

`scale_*_mnirs()` — Colour & fill scales with mnirs palette

`format_hmmss()` — Plot axis in time units

`breaks_timespan()` — Plot axis with time breaks